Latest updates: https://dl.acm.org/doi/10.1145/3755881.3755888

RESEARCH-ARTICLE

A Novel LLM Approach of Cybersecurity Threat Analysis and Response

TIAN HU, Institute of Information Engineering, Beijing, China

SHANGYUAN ZHUANG, Institute of Information Engineering, Beijing, China

ZHAORUI GUO, Institute of Information Engineering, Beijing, China

JIYAN SUN, Institute of Information Engineering, Beijing, China

YINLONG LIU, Institute of Information Engineering, Beijing, China

WEI MA, Institute of Information Engineering, Beijing, China

View all

Open Access Support provided by:

Institute of Information Engineering

Innovation Academy for Microsatellites of CAS

PDF Download
3755881.3755888.pdf
14 January 2026
Total Citations: 0
Total Downloads: 495

Published: 20 June 2025

Citation in BibTeX format

Internetware 2025: the 16th International Conference on Internetware
June 20 - 22, 2025
Trondheim, Norway

Conference Sponsors:
SIGSOFT

Internetware '25: Proceedings of the 16th International Conference on Internetware (June 2025)
https://doi.org/10.1145/3755881.3755888
ISBN: 9798400719264

A Novel LLM Approach of Cybersecurity Threat Analysis and Response

Tian Hu
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China
School of Cyber Security
University of Chinese Academy of Sciences
Beijing, China
jack.mrhu@gmail.com

Shangyuan Zhuang
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China
School of Cyber Security
University of Chinese Academy of Sciences
Beijing, China
zhuangshangyuan@iie.ac.cn

zhaorui Guo
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China
School of Cyber Security
University of Chinese Academy of Sciences
Beijing, China
guozhaorui@iie.ac.cn

Jiyan Sun*
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China
School of Cyber Security
University of Chinese Academy of Sciences
Beijing, China
Sunjiyan@iie.ac.cn

Yinlong Liu
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China
School of Cyber Security
University of Chinese Academy of Sciences
Beijing, China
liuyinlong@iie.ac.cn

Wei Ma
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China
School of Cyber Security
University of Chinese Academy of Sciences
Beijing, China
mawei@iie.ac.cn

Hongchao Wang
Institute of Information Engineering
Chinese Academy of Sciences
Beijing, China
School of Cyber Security
University of Chinese Academy of Sciences
Beijing, China
wanghongchao@iie.ac.cn

lingfeng zhao
Innovation Academy for Microsatellites
Chinese Academy of Sciences
Shanghai, China
zhaolf@microsate.com

xiaojie zhang
Innovation Academy for Microsatellites
Chinese Academy of Sciences
Shanghai, China
zhangxj@microsate.com

Abstract

Satellite-based cloud computing cybersecurity threats have long posed significant challenges, particularly for cloud infrastructure operators. While prior research has partially addressed these issues by mitigating threats and enhancing human response efficiency, this paper proposes a novel AI-Driven Threat Analysis and Response (TAR) framework. The study progresses in three main phases: (1) redefining urgent threats through a novel formula; (2) implementing a triage and analysis framework using augmented Large Language Models (LLMs); and (3) automating incident response via a Security Orchestration, Automation, and Response (SOAR) platform. Our prototype, tested in a simulated public cloud environments using real production threats, demonstrated a 17% improvement in handling low- and medium-urgency threats. Experimental results show our approach achieves 97.8% coverage in automatic threat classification, significantly outperforming traditional manual methods, which achieve 77.8% coverage. With high recall and precision in managing low- and medium-urgency threats, our method enhances manual efficiency through SOAR-enabled automation. Furthermore, our augmented method surpasses the state-of-the-art GPT-4 Turbo model in addressing security threats containing Chinese characters.

*Corresponding Author.

Keywords

Cybersecurity, LLM, Threats Analysis and Response

ACM Reference Format:
Tian Hu, Shangyuan Zhuang, zhaorui Guo, Jiyan Sun, Yinlong Liu, Wei Ma, Hongchao Wang, lingfeng zhao, and xiaojie zhang. 2025. A Novel LLM Approach of Cybersecurity Threat Analysis and Response. In *the 16th International Conference on Internetware (Internetware 2025), June 20–22, 2025, Trondheim, Norway*. ACM, New York, NY, USA, 11 pages. https://doi.org/10.1145/3755881.3755888

1 Introduction

Background and Motivation. The ongoing mass adoption of satellite-based cloud computing has been a key driver of many of the most trans-formative tech trends. With the cloud as the designed primary hosting platform, massive cybersecurity threats have been identified on the cloud. Cybersecurity threats management is a challenge [9, 33, 39, 42]. It is said that 93% of security teams cannot

address all of their security threats in the same day according to a security report [1]. Many organizations lack the resources and staff needed to tackle the growing number of threats to their organization. This inability to handle the volume of threats results in many threats going uninvestigated. And that leaves organizations susceptible to serious attacks.

The unprecedented increase in discovered threats raises new challenges in Threats Risk Management (TRM) . There are more than one million logs or aggregated one hundred incident alerts generated everyday per a medium size companies which have 200-500 elastic cloud serveries with less than 100 domains, according to our survey on the production alerts of the cloud tenants. It is a tremendous workload and impossible mission for a satellite-based cloud computing provider dealing with when it comes to hundreds of tenants. Especially, it is a phenomenon that occurs when cybersecurity professionals are inundated with such a high volume of security alerts that it leads to a diminished ability to react effectively to and investigate real threats. The biggest risk of alert fatigue is that operators will overlook important information. As a fundamental assumption, the alerts can not be eliminated to zero, and the system can not be patched 100%, because of the reality and complexity in productions environments. Hence, we are trying to response the satellite-based cloud computing threats automatically by leveraging the LLMs to reduce the workforce and impove efficiency. The massive threats will be filtered first, then the threats will be prioritized based on urgent score and adjusted by LLMs. finally, the threats will be categorized by LLMs and responded by Security Orchestration, Automation, and Response (SOAR) automatically.

Problem and Challenge. As a result, because of the lack of sufficient manpower, the unprocessed threats remain in the cyberspace which will result in high risk in the future. The phenomenons we mentioned above contain three topics:

First, the current researches [18, 22, 23, 45] focus on a certain aspect, such as alerts compression (threats/alerts filtering, aggregation of threats of the same type, and sampling of the same threats within a short period of time), automatic alerts analysis or alerts response, lacking an end-to-end (from the alert collection, compression, classification and rating as well as automatic response) framework and systematic solutions.

Second, the alerts retrieved from different security products are not uniform [15, 19, 24], the volume of satellite-based cloud alerts increased significantly, hence, it is difficult to prioritize them by leveraging the cloud capabilities rather than intrinsic severity.

Third, due to the continuous dynamic changes of satellite-based cloud threats [12], the current threats types and definitions cannot be accurately defined without combination with the contextual scenarios, resulting in a gradual decrease in the coverage of manual rules and the inability to form an effective automatic response strategy.

In order to prevent alert fatigue in Cybersecurity, we notice that there are several methods [5, 6, 46] mainly focus on the implementation of rules or pattern to identify threats. It is fit for log and event filtering, aggregating, blocking, ranking or prioritizing. However, when it comes to threats generated by different security products and collected from Security Operation Center (SOC) or Security Information and Event Management (SIEM), it is hard to filter and compress because of lack of context and correlation analysis. Moreover, how to remediate and respond the massive threats automatically on the satellite-based cloud is a great concern. Specifically, it is tremendous work to categorize, prioritize as well as respond the threats automatically.

To this end, we propose a novel Automation Method of Cybersecurity Threats Analysis and Response (CTAR) based on Large Language Model Framework which can handle the above problems efficiently and effectively. In order to address the challenges, we divide the methods into three steps: First, retrieve all the different satellite-based cloud native security threats on the cloud from SOC or SIEM. Second, define the threats urgent level, classify and prioritize the threats rules and redefine the score of urgent threats based on five key factors and do threats triage by conducting LLMs. Third, implement the incident response playbook on the SOAR Platform through invoking the cloud native APIs automatically.

Innovation and Contribution. To address the aforementioned challenges effectively, we proposed a novel LLM Threat Analysis and Response (TAR) approach. Detailed discussions follow later in this paper, where the TAR framework is outlined in three main components: Threat Definition, Threat Analysis, and Threat Response. Overall, our approach provided a cost-effective TAR framework with enhanced performance based on our augmented LLM.

Summarily, this paper presented the following key contributions and innovations:

1) ***A novel formula for defining threat urgency.*** We introduced a novel formula to assess the urgency and severity of threats, leveraging existing cloud security mitigation capabilities to refine threat severity and urgency across five critical factors. Experimental results showed a reduction of 17% in high-risk threats, reclassifying them as low or medium risk.

2) ***An end-to-end LLM-based threat analysis and response framework.*** Our comprehensive framework identified, categorized, and rated threat urgency, integrating LLMs to prioritize and analyze threats. The framework achieved 97.8% coverage in automatic threat classification, compared to 77.8% coverage with manual methods, thus significantly reducing the need for manual intervention and enhancing accuracy.

3) ***A cost-effective automated threat response approach using SOAR.*** We proposed a cloud-native SOAR platform that enhances human efficiency by executing predefined playbooks through cloud-native APIs. Our approach significantly outperformed current methods, processing threats more accurately and substantially saving time and effort for security analysts.

2 Related Works

In this section, we first introduce the alerts reduction technologies. Then, we will review the alerts rating technology and automatic alerts triage and response technology.

Reduce the alerts, define the severity, the urgency of the alerts and categorize and respond them automatically and intelligently is very important [25]. Although there are some methods [17, 35, 41] addressed these issues partly, they are difficult to be implemented in practices on the cloud because of lack of automatic end-to-end perspective.

2.1 Alerts reduction technology

Although intrusion detection system (IDS) systems have played a pivotal role in cybersecurity in recent decades [5, 6, 8]. In recent years, tremendous efforts have been devoted into alert retrieving in both academia and industry. In general, alert management techniques can be categorized into three stages: alert preprocessing (alert blocking, filtering, and aggregation), alert analysis (correlation, severity defining, prioritization and ranking) and alert post-processing (prediction, notification and blocking). Chyssler et al. [20] use a static filter with aggregation to reduce false alarm in IP network with the help of end-user involvement to adjust filter rules. Lee [30] present a new semi-supervised and fine-grained filtering method that uses not only alert names and IP addresses but also the payload and semi-supervised clustering results from the alerts, however,they focus on the network intrusion alerts, ignore the application and endpoint level alerts.

Some researchers [16, 26, 29, 43] study new methods to handle the alerts monitor, aggregation, suppression, and make some progress in the fields.

In this paper, we present a reduction method to reduce alert fatigue by conducting alerts urgent score rather than rating the alerts by the alert intrinsic severity.

2.2 Threats rating technology

Security teams today are drowning in alerts. Tracking, analyzing and managing all this alerts manually is challenging, if not impossible. Legacy IDS/IPS tools have failed to keep up with the dramatic changes seen across the modern attack surface, as well as the significant increase in the quantity and diversity of today's cloud alerts. As a result, security and IT teams are buried with more work than they can handle, as they struggle to address all alerts that classifies as "high" or "critical". There are some advanced alerts priority based on machine learning have been proposed to triage in various scenarios[20]. Likewise in [11], the paper proposes a framework for comparing six alert ranking algorithms and identify the best conditions to separate actionable from un-actionable alerts.

Shah et al. [40] addressed the challenge proposing a two-step approach to optimal selection of alerts for investigation methodology that aims: to resolve critical gap between the time needed (demand) and the time available (limited analyst resource) for alert investigation. The paper present a new framework for alert prioritization and selection for further investigation by analysts. A new notion of maximizing cumulative alert risk-score for the investigative workload (minimizing the residual risk-score) is defined in the research. In conclusion, the new two-step framework is a paradigm shift in how alerts are prioritized and selected for investigation because it considers several alert characteristics, resources, and the desired performance metrics in its selection decision-making process.

Some studies [15, 31, 34, 36] move on to introduce proposed method computes the infection score of hosts by modeling, discovery, and analysis of causal relationships among APT steps, each Intrusion Kill Chain (IKC) (simliar to ATT&CK which is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations) is assigned a score of being a part of an APT attack against the corresponding host and finally, a normalized attack surface value is computed for each host to show its rank of exposure to probable APT attacks. Moreover, sachidananda et al. [38] utilizes various attributes and relationships provided by the [7] online scanning engine analysis results to prioritize a set of Indicators of Compromise (IoCs). By concentrating on properly analyzed and ranked IOCs from a vast dataset, the work assists organizations in allocating resources and efforts more effectively. Nevertheless, it is lack of pragmatic method to implement on the cloud computing infrastructure.

2.3 Threats automatic triage and respond technology

Automated correlation and triage technology identifies and groups related alerts to minimize redundancies and allow cybersecurity professionals to focus on the most critical alerts. Hassan et al. [27] presented a method to combat this challenge using contextual and historical information of generated threat alert. The method first generates a causal dependency graph of an alert event. Then, it assigns an anomaly score to each edge in the dependency graph based on the frequency with which related events have happened before in the enterprise. Then it propagates those scores along the neighboring edges of the graph using a novel network diffusion algorithm and generates an aggregate anomaly score which is used for triage. Zhong et al. [48] mines past analysts' operation traces to generate a state machine for automated alert triage. There are many other propose approaches to reduce the number of alerts such as careful configuration and improved classification methods [14, 18]. We also noticed that [13, 16, 32] propose a new alert screening scheme that leverages artificial intelligence (AI)-assisted tools and LLMs to distinguish actual threats from false alarms without investigating every alert. The proposed system framework leverages machine learning and data visualization techniques to analyze alert logs generated from multiple security appliances and distill alerts that are highly critical. How to respond and predict the alerts promptly and automatically is a big problems. [10, 15, 37, 43, 44, 47] However, there are still too many threat alerts for the analysts to properly investigate. In the paper, we try to introduce application's LLM to help us to prioritize the threat alerts, we know that some risks need to be taken care of [3, 21], to the best of our knowledge, our holistic method is the first end-to-end approach. We leverage LLM to automate the alert triage process and response in a cloud native scenario for an enterprise without analysts involvement in handling the low- and medium-risk alerts.

3 Methodology

3.1 Overview

This research addresses a critical gap in existing frameworks: the need for end-to-end automatic threat prioritization and response solution, incorporating alerts handling man-day trade-offs and risk acceptance to support optimal decision-making within resource-constrained cloud environments facing extensive security threats.

As illustrated in Fig. 1, the Threat Analysis and Response (TAR) framework comprises three primary components—Threat Definition, Threat Analysis, and Threat Response—supplemented by a SOAR platform. The methodology proceeds as follows: First, we calculate severity and urgency scores for threats from various sources

like Security Information and Event Management (SIEM) or Security Operations Center (SOC) and redefine the urgency score by the middleware called Manged detection and response (MDR), the score will be fine-tuned by the LLM accordingly. Second, we categorize and analyze threats using LLMs. If a function is defined, the LLM invokes it directly; otherwise, the LLM engages in reasoning, summarizing, and restructuring prompts before invoking the function. Third, we respond to the threats via the SOAR platform, using LLM-generated threat categorization plans to automatically invoke cloud-native API capabilities, thereby hardening cloud infrastructure. If the alerts is tagged as high or critcal, then the alerts tickets will be assigned to the security expert, otherwise, the tickets will be handled automatically by MDR. The following sections describe these components in detail:

3.2 Threat Definition Based on Formula and LLM

To assess the real threat severity in cloud scenarios, we retrieve original threat severity data from the Cloud SIEM or SOC by MDR services via APIs and all the data is stored in real-time database called simple log service which is similar to AWS CloudWatch. In our approach, all scores are defined and calculated by MDR middleware. Irrespective of nomenclature, the scores can be mapped to the qualitative ratings defined in TABLE 1, which outlines the evaluation scale as follows:

Table 1: The Score of the Threat Urgency

Low	Medium	High	Critical
0.1-3.9	4.0-6.9	7.0-8.9	9.0-10

- **Critical Urgency Threats**: Alerts with scores exceeding 9.0 signify urgency and necessitate immediate action to resolve the issue or rapidly mitigate the threat. These alerts typically originate from malicious attackers targeting critical internet assets, are in the mid-to-late stages of intrusion, occur multiple times, or associated with different devices and are not intercepted by current cloud security measures. Additionally, Intruders can easily gain control of the host, which may lead to a threat to the entire network security.
- **High Urgency Threats**: Alerts with scores between 7.0 and 8.9 indicate that these alerts require rapid response and handling to prevent further escalation to critical-level incidents. These alerts typically originate from malicious attackers targeting critical or important Internet assets, are in the mid-to-late stages of intrusion, and are not intercepted by current cloud security measures. Intruders may gain control of the host.
- **Medium Urgency Threats**: Alerts with scores between 4.0 and 6.9 suggest that these alerts may pose a certain threat to the security of systems or networks, but the threat level is relatively moderate. These alerts typically originate from malicious attackers targeting important or general Internet assets, are in the early or mid-stages of intrusion, and are not intercepted by current cloud security measures or similar behaviors may be associated with different devices. Intruders may be able to obtain specific information stored on the host. Medium-priority alerts need to be handled in a timely manner to prevent further escalation to high-priority alerts.
- **Low Urgency Threats**: Alerts with scores between 0 and 4 can be ignored or deferred based on the cost of handling. They may be caused by configuration errors, false detection results, or other non-malicious reasons. These alerts typically originate from non-malicious sources, target general Internet assets or internal assets, are in the early stages of intrusion, and are intercepted by current cloud security measures. Intruders may be able to collect sensitive information from the host.

The common alert dataset is shown in Table 2. And the original alerts have the key attributes, such as the file hash or path, process ID or path, timestamp, command-line or command path, event ID, source IP or target ID, container or server, and more. We propose a novel formula for determining threat urgency, which incorporates the principles of Attack Source Score, Attack Target Score, Attack Phase Score, Threat State Score, and Attack Correlation Score. This formula allows cloud tenants to evaluate and customize the urgency of each threat according to their specific risk appetite.

The urgency score for addressing a threat dynamically changes, as illustrated in Fig. 2. The threat urgency rating system which is part of the MDR middleware operated by the cloud providers calculates the overall score based on five key indicators: Attack Source, Attack Target, Attack Phase, Threat State, and Attack Correlation.

We provide the following formula to calculate the score of urgency to respond a threat. Equations are defined below:

$$Si = \sum_{z=1}^{z=3} f_z(xi) - f_4(x_i) + f5(x_i) \tag{1}$$

where

- S_i: the overall threat score of threat number i; $S_i \in \{0, 10\}$. If it is negative, take the value as 0; If the score is bigger than 10, take the value as 10;
- $f_1(x_i)$: Attack Source Score of threat nubmer i;
- $f_2(x_i)$: Attack Target Score of threat nubmer i;
- $f_3(x_i)$: Attack Phase Score of threat nubmer i;
- $f_4(x_i)$: Attack State Score of threat nubmer i;
- $f_5(x_i)$: Attack Correlation Score of threat nubmer i.

The overall score range: 0-10, the elements in the formula can be described as following:

- **Attack Source Score.** In most cases, this attribute corresponds to IP addresses. The values are retrieved from various security vendors(Alien Vault OTX, Spider foot, Mitaka), and the mean score of the IP reputation field values is calculated. Valid scores range from 0 to 2, with a score of 2 indicating a malicious reputation and a score of 0 indicating trust. The score originates from TI security vendors. And we apply the function $f_1(x_i)$ to the IP reputation, Finally $f_1(x_i)$ can be expressed as:

$$f1(xi) = \sum_{i=0}^{z=n} f1(x)/n \in \{0, 1, 2\} \tag{2}$$

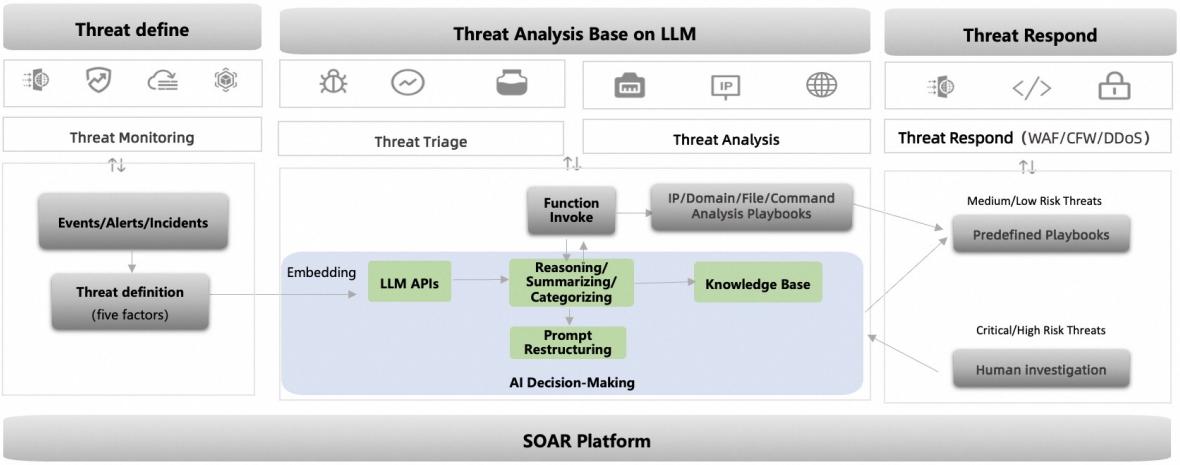

Figure 1: The Framework of the Automatic Threats Analysis and Response(TAR) based on LLM

Table 2: Common Cloud Security Alerts

Alert Name	Alert Category	Original Alert Level
Malicious DNS Request Interception	IP/Domain Related Alert	Medium
Cloud Security Center Client Abnormal Offline	Other Category	High
Log/Image File Containing WEBSHELL Code	Malicious File	Critical
RAM Sub-account Login from Different Location	IP/Domain Related Alert	High
Detection of Backdoor (Webshell) File	Malicious File	Critical
Suspicious OSS Access Behavior	IP/Domain Related Alert	Suspicious
ECS Login from Unusual Location	IP/Domain Related Alert	Medium
Malicious Tampering of Client Files	Other Category	Critical
ECS Login from Unusual IP	IP/Domain Related Alert	Suspicious
Sensitive Registry Item Protection	Malicious Command Behavior	Critical

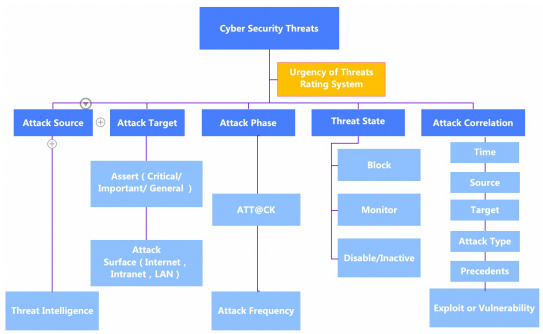

Figure 2: The Urgency of Threats Rating System Components Breakdown

- **Attack Target Score.** The Attack Target Score depends on the asset value denoted as $a(x_i)$ and the attack surfaces score which can be expressed as $b(x_i)$. The values are defined as below: 1) Critical assets score has a score of 2, important asset score of 1, while the general asset score of 0; 2) The attack surface is internet-facing which has a score of 2, and an intranet asset a score of 1, a LAN asset a score of 0; Thus, a critical internet-facing asset would have a combined score of 4. The score originates from cloud security product.

$$f_2(x_i) = a(x_i) + b(x_i) \in \{0, 1, 2, 3, 4\} \tag{3}$$

- **Attack Phase Score.** The Attack Phase score depends on the value of the kill chain which can be expressed as $c(xi)$ and the attack frequency score which can be denoted as $d(xi)$. we notice that [4], cyber skill chain is part of the intelligence driven defense model for identification and prevention of cyber intrusions activity and the seven steps of the cyber kill chain enhance visibility into an attack and enrich an analyst's understanding of an adversary's tactics, techniques and procedures. We apply the 14 steps phrase to estimate the score according to ATT&CK [2]. Step 1-3 for the pre-attack, have a score of 0, step 4-7 of during-attack have a score of 1, and step 8-9 of post-attack have a score of 2. On the other hand, the attack frequency score reflects historical attack vectors used by the intruder, with valid values ranging from 1 to 2. If a threat is triggered more than once or involves two or more products, it is assigned a score of 2; otherwise, it receives a score of 1. This score originates from cloud security product.

$$f_3(x_i) = c(x_i) * d(x_i) \in \{0, 1, 2, 3, 4\} \tag{4}$$

- **Attack State Score.** The response action depends on the following classifications: 1) Block: Value of 10, indicating

the threat does not require further handling because of the alerts was blocked by cloud security products; 2) Monitor: Value of 1, indicating the threat requires active monitoring; 3) Disable or inactive: Value of 0, indicating the threat should be addressed, enabled, and filtered on the cloud security products. This score originates from cloud security product.

$$f_4(x_i) \in \{10, 1, 0\} \tag{5}$$

- **Attack Correlation Score.** The score depends on the following criteria: if any two or more attack correlation factors (e.g., time, source, target) match, the score is 1; otherwise, it is 0. This score originates from cloud security product.

$$f_5(x_i) \in \{0, 1\}$$

After retrieving the original score, we input the redefined score as a prompt to our LLM. Organizations require an approach to address emerging threat scenarios by prioritizing threats, resources, and actions. This need is met by the use of pre-trained datasets and augmented LLMs (which are based on the cloud security experts empirical practise and feedback from the investigation of each specific alert). The LLM implemented in our work is based on the fundamental transformer large language model which is a 72B-parameter version of the large language model series of Qwen (abbr. Tongyi Qianwen). Qwen-72B supports 32k context length, which is pretrained on over 3 trillion tokens, including Chinese, English, multilingual texts, books, codes, and mathematics, covering general and professional fields, including commonsense, reasoning, code, mathematics, etc.

3.3 Threat Triage and Analysis Using LLM

We propose a method for embedding threats into the Qwen-72B model, enabling it to decide whether to invoke a predefined function or consult an LLM for reasoning, summarizing, categorizing, and restructuring prompts. All actions are logged and stored in a knowledge base for future reference. The architecture of this threat triage and response scheme is shown in Fig. 3. Threats retrieved from the SIEM or SOC are categorized into four main groups by the LLM. The SOAR platform's predefined playbook script is then triggered to respond to the threats accordingly, allowing the AI to handle a large volume of potential security threats automatically. All the outputs by LLMs are tagged into AI and function call. The output generated by AI means the low-and medium-risk alerts can be executed by invoking cloud APIs becuase of the high F1 score. Meanwhile, the high- and critcal-risk alert related tickets need to be reviewed by the security expert on the MDR middleware platform. On the other hand, the output is tagged with function call means the alerts can be executed by invoking function call with high accuracy. Eventually, users will know which outputs are executed by AI or function call. We can enhance transparency and facilitate the AI process.

As shown in Fig. 4, different type of threats are interconnected. This section describes the correlation of threats in the context of an actual attack scenario where an attacker successfully compromises a web system through brute-force attacks, subsequently uploads a backdoor to the target host, and then engages in lateral movement. Initially, this chapter obtains raw alert information generated by various cloud services from a SIEM or SOC . For instance, Web

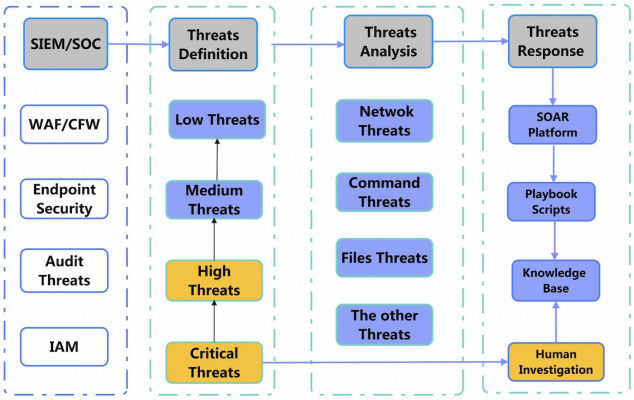

Figure 3: The Threat Implementation Architecture

Application Firewall (WAF) or Firewall (FW) detects and identifies a large number of brute-force attack alerts. Upon successful exploitation by the attacker, this triggers endpoint detection and response (EDR) alerts for the upload of a Trojan file on a specific server. The corresponding Trojan family can be identified through threat intelligence based on the file hash. The reputation analysis of the source IP can also quickly determine the historical context of the source IP. When the attacker engages in further malicious activities such as privilege escalation or lateral movement, generating command or process alerts, these alerts can be correlated to the corresponding trojan file or server location. Thus, it is determined that this is an attack based on successful brute-force attacks followed by the upload of a trojan.

All the alerts collected by the cloud SIEM or SOC are categorized into four main groups by the LLM: the IP/Domain related network Threats, the endpoint Malicious Command Threats, the endpoint Malicious Files Threats and The other Threats.

The four main group Threats can be described as follows:

- **Malicious IP/Domain**. The attack or audit alerts contain any IP or domain related record such as: the source or target IP addresses, login IP address, outbound traffic, ports or Threat Intelligence, etc.
- **Malicious Commands**. The chain of process, the command line, the thread, the path of command, the services ID, ports, the hash of process, etc.
- **Malicious Files**. The hash of files, the path of the files, the name of the files, the container files, web parser, the cyber threat intelligence of the files, etc.
- **The Others**. The zero day attack, the APTs or the latest variant of attacks. In most cases, the alerts categorized into The Others can not be responded automatically and require human intervention.

After we defined the score of the threats, the threats were embedded to the LLM's API ,then the prompt is summarized as a input as follows:

Instructions:

You are a Senior Security Operations Specialist in MDR team. Your task is to assist the security expert in analyzing and drawing

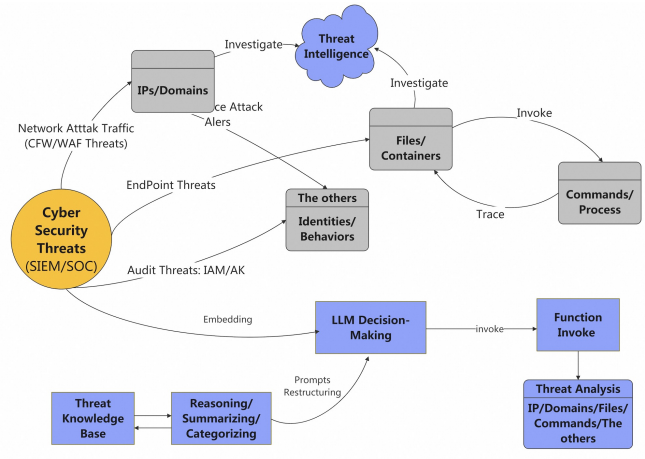

Figure 4: The Threat Analysis Approach

conclusions on security threats based on the threat details. Your analysis should include (but not limited to) the following:

- Threat level of the security incident
- The conclusion of the security threats
- Reasoning process of whether the security threats has a real risk or not
- Recommendations for handling the security threats
- Please use simple language in less than 100 words, wherever possible.

Facts:

Threats level: the security threat urgency score is defined as: the highest critical severity is 10.0 points and the lowest is 0.0 points.

Conclusion: an expert conclusion is given comprehensively.

Analysis process: describes the root cause process of the conclusion of the whole security event from the perspective of security experts.

Suggestion: give professional security suggestions from the security expert's point of view.

Policy:

The threat level takes precedence with reference to the following rules:

- For maliciously corrupted client file threats, if the process is one of perf, /usr/bin/vi, /usr/bin/vim, or dockerd, the alert level is false alarms.
- Unless other security events have occurred recently on the server, the alert level for both the agent offline and malicious destruction of client threats levels are marked as Low.
- Alert level for client offline alert is low risk if the number of recent offline alert is greater than 1.
- Alerts level for mining programs, bouncing shells, hacking tools that use access key, ransomware, and portal services that perform suspicious sequences of behaviour are all flagged as critical.
-Two types of alerts in terms of service account creation of pod, K8s Bind User to administrator role marked as low risk, unless other different types of security incidents have occurred recently.
- When Risky docker remote debugging interface alerts occurs, alert level is marked as medium.

Respond:

Please make an analysis based on the above requirements. You should strictly follow the following format when outputting respond without any extra characters:

$$\begin{aligned} \text{“threats_level"} &: \text{“}highrisk\text{”}, \text{“}score\text{”} \\ \text{“Conclusion"} &: \text{“}xxxx\text{”} \\ \text{“Suggestion"} &: \text{“}xxxx\text{”} \\ \text{“Alarm information"} &: \{inputdata.datalist.*.threatsinfo\} \\ \text{“Process details"} &: \{event.event_detail\} \end{aligned}$$

We use the datasets of more than five hundred thousand desensitised threats retrieved from the cloud tenants' practical production environments. The datasets is to pretrain our augmented model based on Qwen-72B model to categorize the alerts and then define the threats urgency score according to the cybersecurity experts rating equation. All the threats are prioritized and redefined by LLM and trigger the SOAR playbook script to respond the threats automatically.

3.4 Threat Automatic Response using SOAR

Inspired by the work of Hu et al.[28], we implement and escalate threats based on their categories and the score of urgency according to the scheme of the LLM based on augmented datasets. The low- and medium-risk threats are responded according to the playbook scripts of the SOAR platform by invoking the cloud mitigation APIs.

The operating mechanism of the threats automatic response scheme is shown in Fig. 5. Specifically, the workflow in our scheme goes like this: 1) retrieve all the threats from cloud SIEM or SOC according to prior define of urgency score of the threats. 2) If it is a layer 4-7 threats, check whether the tenant has the CFW in place and whether the CFW has a virtual patch, if not, then check the CFW policy, adjust and enable the strict mode. 3) If it is a layer 7 threat, for example, an application SQL injection threat, check whether the tenant has the WAF in place and whether the WAF has a virtual patch, if not, then check RASP feature, adjust and enable the strict mode. 4) If it is a endpoint threat, check whether there is a host virtual patch in places, if not, whether the threat is related designated files ,process or IPs, then whitelist it. if not , go to the host level countermeasures, such as kill the malicious process, delete the malicious files or add security group to isolated the endpoint. 5) End until the policy is automatic configured. All the response process goes according to the assigned playbook script by invoking the cloud native APIs. At last, according to the playbook script on the SOAR based on the categories of threats, Those threats with low F1 score need to be investigated by human.

Because the holistic playbook is complex, we use the Web Application Firewall (WAF) as an example, noting that other cloud-native security products follow a similar approach to threat handling. Establishing an automatic remediation process involves three steps: First, input the cloud tenant ID to retrieve all threats associated with that ID. Second, input all retrieved threats to obtain the remediation plan. Third, execute the WAF's remediation playbook script to ensure that the identified threats are protected by the WAF. The playbook script operates as follows:

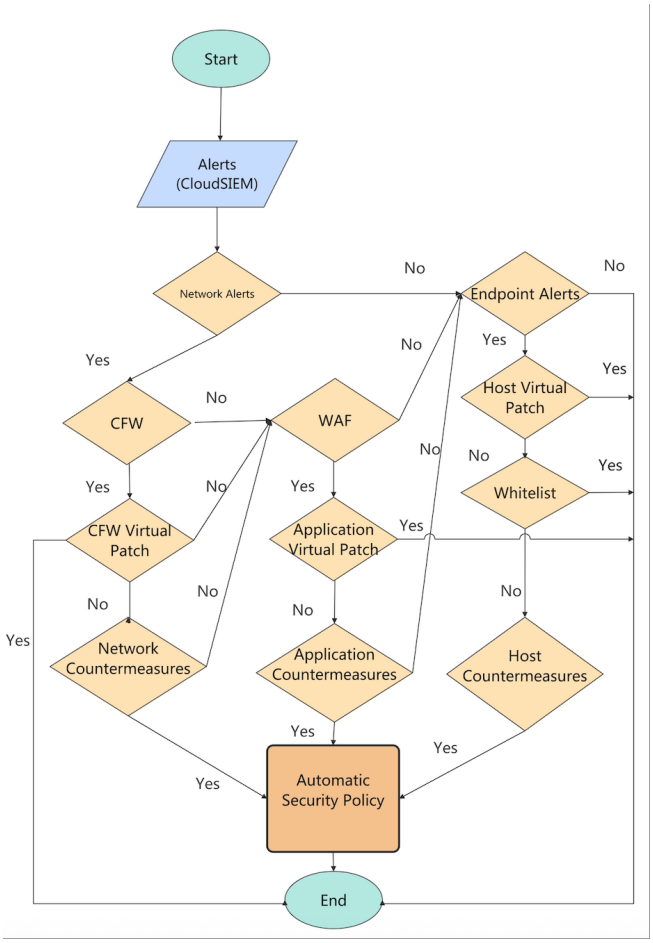

Figure 5: The threats response scheme workflow

Algorithm 1 The WAF Playbook Script

Require: threats v from user ID uid, downgrading conclusions con

```
if match v and con == 'advanced configuration' then
    Remediate threats by loose mode by invoking API
else if match v and con == 'strict mode' then
    Remediate threats by strict mode by invoking API
else if match v and con == 'enable or upgrade' then
    Update the urgency of threat v
else if match v and con == 'green' then
    Neglect threat v
else
    Other remediate method
end if
```

4 Experiments and Results

In this section, we conduct experiments to evaluate the performance of our proposed methodology. First, we assess the effectiveness of threat definition and triage using our augmented LLM model. Second, we verify the coverage of threats identified by our augmented LLM in comparison to manual analysis. Furthermore, we conduct experiments to compare the F1 score of our method with that of the current GPT-4 Turbo model.

4.1 Experimental Settings

In order to demonstrate the real scenario, we retrieve a real environment dataset of 1000 typical threats from independent random cloud tenants' cloud SIEM. We implement an augmented transformer-based LLMs, specifically the Qwen-72B model, adjusting the prompts with a knowledge base in Chinese.

4.2 Baseline

First, we test the downgrade of our urgency of threat definition with the origin threats. Second, we compare the coverage of the threat of our LLM model and non LLM models. Lastly, we compare the urgency scores of threat triage rates with those generated by the current GPT-4 Transformer model. Thus, we employ our approach to assess urgency scoring system and the effectiveness of threat categorization respectively. The ground truth labels of the alerts are obtained from SIEM and granted and reviewed by security experts on the MDR middleware.

4.3 Evaluation Metrics

To evaluate the efficiency of the LLM methods, and drawing inspiration from classical metrics in the field of machine learning, we assess triage performance using the following metrics:

- **Accuracy**= $\frac{TP+TN}{TP+TN+FP+FN}$
- **Precision**= $\frac{TP}{TP+FP}$
- **Recall**= $\frac{TP}{TP+FN}$
- **F1**= $2 \times \frac{Precision \times Recall}{Precision+Recall}$

4.4 Threat Urgency Definition Evaluation

We evaluate the percentage of threat recognition based on threat urgency by retrieving 1000 original threats categorized as critical, high, medium, and low risk from the SIEM. Each phase of threats including the orginal (generated by security products), equation (generated by our definition) , LLM (adjusted following the equation) are carefully reviewed by security expert. As shown in TABLE 3, our equation method identifies 64.5% of threats as low- and medium-risk, representing an increase of 14.5% by the equation. Additionally, the proportion of critical risk threats is reduced by 15% through the equation, indicating that these threats can be downgraded due to existing remediation capabilities. Furthermore, as shown in Fig. 6, the LLM adjusts the total classification to 67% for low and medium risk threats. In conclusion, our approach demonstrates an improved real threat urgency rate when implemented in cloud environments.

4.5 Threat Automatic Triage and Response Evaluation

To evaluate the effectiveness of the LLM, we compare the threat triage rate of the LLM with that of manual analysis. As shown in TABLE 4, our method achieves an average threat recognition rate of 97.8%, compared to 77.8% for manual labor. All categories of threats, including IP/domain threats, file threats, and command

Table 3: Threat Urgency Recognition

Threat Categories	Original	Equation	Urgency by LLM
Low Threat	250/25%	435/43.5%	450/45%
Medium Threat	250/25%	210/21%	220/22%
High Threat	250/25%	255/25.5%	200/20%
Critical Threat	250/25%	100/10%	130/13%
Recognition	500/50%	645/64.5%	670/67%

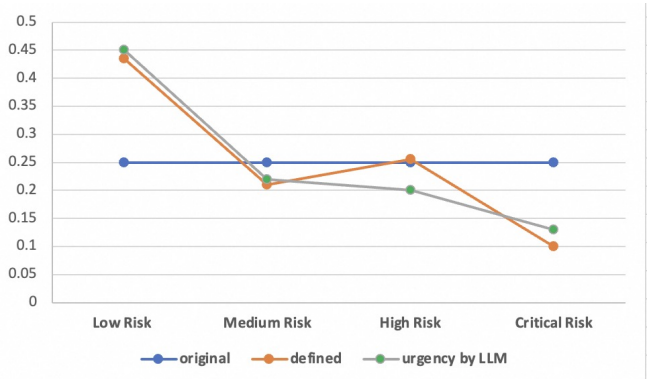

Figure 6: Threat urgency by LLM

threats, demonstrate high recognition rates. Additionally, the manual methods fail to identify 22.2% of threats due to human resource limitations. On the contrary, our LLMs based method can identify those threats automatically without pre-defined attack pattern called The Others threats effectively.

Table 4: Threat Category Recognition Percentage by LLM

Threat Categories	Total	Manual	Ours
IP/domain Related Threat	567	300/52.91%	**565/99.65%**
Malicious Files Threat	320	425/64.75%	**312/97.50%**
Malicious Command Threat	80	53/66.25%	**71/88.75%**
The Others	33	222/22.2%	**30/90.91%**
Total	1000	778/77.8%	**978/97.8%**

In order to evaluate the effectiveness of the LLM when it comes to threat severity on the Cloud by our integrated method. We retrieve the 1000 threat datasets from cloud production environments. As shown in TABLE.5. The severity of threat redefined by ours LLM models as following:

4.6 Performance of Different Models Evaluation

To evaluate the effectiveness and performance of our augmented LLM in assessing the real urgency and severity of threats in the cloud, we retrieve a dataset of 1000 threats from a cloud production environments. Additionally, we introduce the state-of-the-art GPT-4 Turbo model for comparison.

Table 5: threat severity by our augmented models

Category	Accuracy	Precision	Recall	F1
Low threat	100.00%	100.00%	100.00%	100.00%
Medium threat	87.50%	87.50%	100.00%	93.33%
High threat	72.73%	72.73%	100.00%	84.21%
Critical threat	71.62%	71.62%	100.00%	83.46%

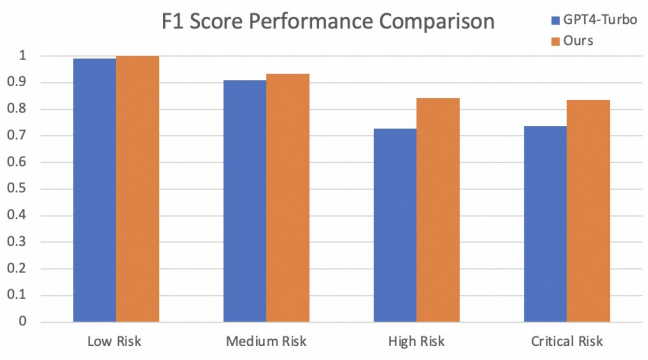

Figure 7: Performance Comparison of Different Models

As shown in Fig. 7, our approach achieves a 100% F1 score for low-risk threats and a 93.33% F1 score for medium-risk threats. The GPT-4 Turbo model also demonstrates high performance, attaining an F1 score of 99.16% for low-risk threats and 90.91% for medium-risk threats. Unfortunately, both our method and the current GPT model exhibit significantly lower F1 scores for high and critical threats. As a result, high and critical threats require human intervention for further investigation.

4.7 System Evaluation

In conclusion, our method enables us to achieve a nearly optimal scheme for threat analysis and response. Experimental results demonstrate an increase of 17% in the identification of low and medium-risk threats, with our approach covering 97.8% of threats, compared to only 77.8% for traditional manual methods. Our recognition rate for low and medium-urgency threats reaches an F1 score higher than 93.3% for automatic responses based on the SOAR platform, outperforming the current GPT-4 Turbo model. Ultimately, we can automate responses to low and medium-urgency threats while reserving high and critical threats for human investigation with prudence.

5 Conclusion

In this paper, a novel approach of cybersecurity threat analysis and response method(TAR) based on LLMs is proposed. This integrated framework leverages the capabilities of LLMs and cloud native to streamline incident response processes within cloud computing environments, significantly reducing the time and effort required by security analysts compared to existing best practices. In summary,

1) We provide a near-optimal end-to-end framework to identify and rate the score of urgency of threats, which combines LLM to determine the priorities and respond the threats.

2) We propose a novel scheme of cloud security capabilities by leveraging the augmented LLM from the commercial market to prioritize the threats automatically, which improves the overall alarm coverage with high accuracy and efficiency compared with man labors.

3) We implement and respond the threats automatically by leveraging the cloud native Security Orchestration, Automation and Response (SOAR) platform through invoking the cloud native APIs.

Our method is still worth improving in the future. First, the limitation of the approach is based on the LLM, the Security [3] of LLM and the output of LLM should review carefully. Moreover, how to handle the high and critical urgent threats automatically should take into consideration and put it into threats management scope, furthermore we need to improve the recognition rate and accuracy regarding the high and critical threats so that we can respond it automatically. Lastly, the method is only applicable to cloud native infrastructure based on the SOAR, the other scenarios like hybird or on-premises, the adaptations (evaluate the security products' APIs and security capabilities) and extensions (SOAR should support the specific APIs and the alerts need to be collected by SIEM) should be discussed in the future.

References

[1] [n. d.]. 2020 State of SecOps and Automation report. https://www.sumologic.com/blog/2020-state-of-secops-automation-report/.

[2] [n. d.]. MITRE ATT CK is a globally-accessible knowledge base of adversary tactics and techniques based on real-world observations. https://attack.mitre.org/.

[3] [n. d.]. OpenAI. https://https://openai.com/safety.

[4] [n. d.]. The seven steps of the Cyber Kill Chain is developed by Lockheed Martin, the Cyber Kill Chain® framework is part of the Intelligence Driven Defense® model for identification and prevention of cyber intrusions activity. https://www.lockheedmartin.com/en-us/capabilities/cyber/cyber-kill-chain.html.

[5] [n. d.]. Snort: Network Intrusion Detection and Prevention System. https://www.snort.org/.

[6] [n. d.]. Suircata: An open sourced IDS system. https://suricata-ids.org/.

[7] [n. d.]. virustotal: Analyse suspicious files, domains, IPs and URLs to detect malware and other breaches, automatically share them with the security community. https://www.virustotal.com/gui/home/upload.

[8] [n. d.]. Zeek: An Open Source Network Security Monitoring Tool. https://www.zeek.org/.

[9] Radwan Tarek Abdelbaki Nashwa and Azer Marianne A. 2017. Cloud computing security: challenges and future trends. *International Journal of Computer Applications in Technology* 55, 2 (January 2017), 158–158.

[10] H. Albsheer et al. 2022. Cyber-attack prediction based on network intrusion detection systems for alert correlation techniques: a survey. *Sensors* 22, 4 (2022), 1494.

[11] Simon Allier, Andre Hora, Nicolas Anquetil, and Stéphane Ducasse. 2012. A Framework to Compare Alert Ranking Algorithms. In *19th Working Conference on Reverse Engineering*. 277–285.

[12] S. H. Alrasheed, M. Aied Alhariri, S. A. Adubaykhi, and S. El Khediri. 2022. Cloud Computing Security and Challenges: Issues, Threats, and Solutions. In *2022 5th Conference on Cloud and Internet of Things (CIoT)* (Marrakech, Morocco). 166–172. doi:10.1109/CIoT53061.2022.9766571

[13] Tao Ban, Ndichu Samuel, Takeshi Takahashi, and Daisuke Inoue. 2021. Combat Security Alert Fatigue with AI-Assisted Techniques. In *Cyber Security Experimentation and TestWorkshop*. ACM.

[14] D. Barbar´a and S. Jajodia. 2002. Applications of data mining in computer security. (2002).

[15] S. Benferhat, A. Boudjelida, K. Tabia, and H. Drias. 2013. An intrusion detection and alert correlation approach based on revising probabilistic classifiers using expert knowledge. *Applied Intelligence* 38, 15 (2013), 520–540.

[16] K. Bhukar, H. Kumar, R. Mahindru, R. Arora, S. Nagar, P. Aggarwal, and A. Paradkar. 2023. Dynamic Alert Suppression Policy for Noise Reduction in AIOps. (2023), 178–188.

[17] Mainak Chakraborty and Ajit Pratap Kundan. 2021. Architecture of a Modern Monitoring System. In *Monitoring Cloud-Native Applications: Lead Agile Operations Confidently Using Open Source Software*. Springer, 55–96.

[18] Junjie Chen, Xiaoting He, Qingwei Lin, Yong Xu, Hongyu Zhang, Dan Hao, and Feng et al. Gao. 2019. An empirical investigation of incident triage for online service systems. In *ICSE: SEIP*. IEEE Press, 111–120.

[19] Junjie Chen, Xiaoting He, Qingwei Lin, Hongyu Zhang, Dan Hao, Feng Gao, Zhangwei Xu, Dang Yingnong, and Dongmei Zhang. 2019. Continuous Incident Triage for Large-Scale Online Service Systems. In *2019 34st IEEE/ACM International Conference on Automated Software Engineering (ASE)*. IEEE.

[20] T. Chyssler, S. Burschka, M. Semling, T. Lingvall, and K. Burbeck. 2004. Alarm reduction and correlation in intrusion detection systems. In *Detection of Intrusions and Malware Vulnerability Assessment*. Computer Science.

[21] Tianyu Cui, Yanling Wang, Chuanpu Fu, Yong Xiao, Sijia Li, Xinhao Deng, Yunpeng Liu, Qinglin Zhang, Ziyi Qiu, Peiyang Li, Zhixing Tan, Junwu Xiong, Xinyu Kongand Zujie Wen, Ke Xu, and Qi Li. 2024. Risk Taxonomy, Mitigation, and Assessment Benchmarks of Large Language Model Systems. In *arXiv*. arXiv.

[22] Min Du, Feifei Li, Guineng Zheng, and Vivek Srikumar. 2017. Deeplog: Anomaly detection and diagnosis from system logs through deep learning. In *Proceedings of the 2017 ACM SIGSAC conference on computer and communications security*. 1285–1298.

[23] Patel Ahmed et al. 2013. An intrusion detection and prevention system in cloud computing: A systematic review. *Journal of network and computer applications* 36, 1 (2013), 25–41.

[24] S. Ghosh, K. Grover, J. Wong, C. Bansal, R. Namineni, M. Verma, and S. Rajmohan. 2024. Dependency Aware Incident Linking in Large Cloud Systems. In *Companion Proceedings of the ACM on Web Conference*. 141–150.

[25] A. W. Green, A. B. Woszczynski, K. Dodson, and P. Easton. 2020. Responding to Cybersecurity Challenges: Securing Vulnerable U.S. Emergency Alert Systems. *Communications of the Association for Information Systems* 46 (2020).

[26] Dipanwita Guhathakurta, Pooja Aggarwal, Seema Nagar, Rohan Arora, and Bing Zhou. 2022. Utilizing Persistence for Post Facto Suppression of Invalid Anomalies Using System Logs.

[27] Wajih Ul Hassan, Shengjian Guoz, Ding Li, Zhengzhang Chen, Kangkook Jee, Zhichun Li, and Adam Bates. 2019. NODOZE: Combatting Threat Alert Fatigue with Automated Provenance Triage. In *NDSS*.

[28] Tian Hu, Shangyuan Zhuang, Jiyan Sun, Yinlong Liu, Wei Ma, and Hongchao Wang. 2023. A Cost-effective Automation Method of Massive Vulnerabilities Analysis and Remediation Based on Cloud Native. In *2023 IEEE 22nd International Conference on Trust, Security and Privacy in Computing and Communications*. IEEE., 772–779.

[29] Alexis Huet, Jose Manuel Navarro, and Dario Rossi. 2022. Local evaluation of time series anomaly detection algorithms. (2022), 635–645.

[30] Yoonho Lee Hyeon gy Shon and MyungKeun Yoon. 2023. Semi-Supervised Alert Filtering for Network Security. In *Electronics*. MDPI.

[31] Guofei Jiang, Haifeng Chen, Kenji Yoshihira, and Akhilesh Saxena. 2011. Ranking the importance of alerts for problem determination in large computer systems. *Cluster Computing* 14, 3 (2011), 213–227.

[32] Y. Jiang, C. Zhang, S. He, Z. Yang, M. Ma, S. Qin, Y. Kang, Y. Dang, S. Rajmohan, Q. Lin, and D. Zhang. 2023. Xpert: Empowering Incident Management with Query Recommendations via Large Language Models. In *Proceedings of the IEEE/ACM 46th International Conference on Software Engineering*. 1–13.

[33] I. Kanwal, H. Shafi, S. Memon, and M.H. Shah. 2021. Cloud Computing Security Challenges: A Review. In *Cybersecurity, Privacy and Freedom Protection in the Connected World*. Springer, Cham.

[34] Mehran khosravi and Behrouz Tork Landai. 2020. Alerts Correlation and Causal Analysis for APT Based Cyber Attack Detection. In *Computer Science*. IEEE, 162642–162656.

[35] J. Kuang, J. Liu, J. Huang, R. Zhong, J. Gu, L. Yu, R. Tan, Z. Yang, and M. Lyu. 2023. Knowledge-aware Alert Aggregation in Large-scale Cloud Systems: a Hybrid Approach. In *Proceedings of the 46th International Conference on Software Engineering: Software Engineering in Practice*. 369–380.

[36] Derek Lin, Rashmi Raghu, Vivek Ramamurthy, Jin Yu, and et al. 2014. Unveiling clusters of events for alert and incident management in large-scale enterprise IT. In *SIGKDD*. ACM, 1630–1639.

[37] Khoi Khac Nguyen, Dinh Thai Hoang, Dusit Niyato, Ping Wang, Diep Nguyen, and Eryk Dutkiewicz. 2018. Cyberattack detection in mobile cloud computing: A deep learning approach. In *IEEE Wireless Communications and Networking Conference (WCNC)*. 1–6.

[38] Vinay Sachidananda, Bryson Teo, Ruitao Feng, Kwok-Yan Lam, and Liu Yang. 2023. FOCUS: Fatigue-reducing IOCs through Prioritization and Scoring. In *IEEE DASC/CyberSciTech/PICom/CBDCom 2023*. IEEE, 20–27.

[39] S. Sengupta, V. Kaulgud, and V. S. Sharma. 2011. Cloud Computing Security–Trends and Research Directions. In *2011 IEEE World Congress on Services* (Washington, DC, USA). 524–531.

[40] Ankit Shah, Rajesh Ganesan, and Sushil Jajodia. 2019. A Two-Step Approach to Optimal Selection of Alerts for Investigation in a CSOC. In *IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY*, Vol. VOL. 14, NO. 7, JULY. IEEE,

1857–1870.

[41] P. Srinivas, F. Husain, A. Parayil, A. Choure, C. Bansal, and S. Rajmohan. 2023. Intelligent Monitoring Framework for Cloud Services: A Data-Driven Approach. In *Proceedings of the 46th International Conference on Software Engineering: Software Engineering in Practice*. 381–391.

[42] H. Tabrizchi and M. Kuchaki Rafsanjani. 2020. A survey on security challenges in cloud computing: issues, threats, and solutions. *J. Supercomput.* 76, 12 (2020), 9493–9532.

[43] Haiping Wang, Binbin Li, Tianning Zang, Yifei Yang, Zisen Qi, Siyu Jia, and Yu Ding. 2023. Real-Time Aggregation for Massive Alerts Based on Dynamic Attack Granularity Graph. (2023), 225–243.

[44] W. Wang, J. Chen, L. Yang, H. Zhang, and Z. Wang. 2023. Understanding and predicting incident mitigation time. *Information and Software Technology* 155 (2023), C. Online publication date: 1-Mar-2023.

[45] Jingmin Xu, Yuan Wang, Pengfei Chen, and Ping Wang. 2017. Lightweight and Adaptive Service API Performance Monitoring in Highly Dynamic Cloud Environment. In *SCC*. IEEE, 35–43.

[46] Ben Charhi Youssef, Bendriss Elmehdi, Mannane Nada, and Regragui Boubker. 2018. Implementation Approach for IDS based on Risk Assessment and Attack Pattern in Cloud Computing. In *International Conference on Control, Automation and Diagnosis (ICCAD)*. 1–5.

[47] N. Zhao et al. 2020. Understanding and handling alert storm for online service systems. In *ICSE-SEIP*. ACM, 162–171.

[48] C. Zhong, P. Liu J. Yen, and R. F. Erbacher. 2016. Automate cybersecurity data triage by leveraging human analysts' cognitive process. In *IEEE BigDataSecurity*.